Question	Answer	Marks	Partial Marks
2(b)(i)	$(5x-4)(3x+2)$ oe final answer	**2**	**B1** for $(ax+b)(cx+d)$ where either $ac = 15$ <u>and</u> $bd = -8$ or $ad + bc = -2$ or $5x(3x + 2) - 4(3x + 2)$ or $3x(5x - 4) + 2(5x - 4)$ or correct factors seen and spoiled
2(b)(ii)	$\frac{4}{5}$ oe and $-\frac{2}{3}$ oe	**1**	**FT** a factorised quadratic
2(c)	$x(x+4y)(x-4y)$ final answer	**3**	**B2** for $(x^2+4xy)(x-4y)$ or $(x+4y)(x^2-4xy)$ or answer in the form $x(a + b)(a - b)$ or correct answer seen and spoiled or **B1** for $x(x^2-16y^2)$ oe or $(x+4y)(x-4y)$
2(d)	$\frac{1-2a}{x}$ oe final answer	**4**	**B2** for $(2x - 1)(1 - 2a)$ oe or **B1** for $2x - 1 - 2a(2x - 1)$ or $2x(1 - 2a) - (1 - 2a)$ **B1** for $x(2x - 1)$
3(a)(i)	4	**1**	
3(a)(ii)	7	**1**	
3(a)(iii)	8	**1**	
3(b)(i)	14	**1**	
3(b)(ii)	4	**2**	**B1** for [l.q. =] 11 or [u.q =] 15
3(c)	8.09	**3**	**M1** for $5 \times 3 + 10 \times 6 + 43 \times 7 + 75 \times 8 + 48 \times 9 + 21 \times 10$ **M1 dep** ÷ 200
3(d)	30, 70, 40, 36, 24 seen	**B2**	**B1** for 3 or 4 correct or **M1** for $1 \times (80 - 50)$, $3.5 \times (100 - 80)$, $4 \times (110 - 100)$, $3.6 \times (120 - 110)$ and $0.6 \times (160 - 120)$ oe
	(*their* 30×65 + *their* 70×90 + *their* 40×105 + *their* 36×115 + *their* 24×140) ÷ 200	**M3**	**M1** for midpoints soi **M1** for Σfx, x in interval or boundary of interval **M1 dep on second M1** for ÷ 200
	99.75	**A1**	
4(a)(i)	(2, 7)	**2**	**B1** for each coordinate

Question	Answer	Marks	Partial Marks
4(a)(ii)	$-\frac{1}{2}x+8$ oe	**4**	Correct equivalent in different form scores 3 marks. **M1** for gradient of $AB = \frac{9-5}{3-1}$ or $\frac{4}{2}$ or 2 **M1 dep** for gradient $p = -\frac{1}{\textit{their grad of AB}}$ **M1 (dep** on previous M1) for substitution of *their* midpoint into $y = (\textit{their } p)x + c$ oe where *their* $p \neq 0$
4(b)(i)	$\begin{pmatrix}0\\2\end{pmatrix}$	**2**	**B1** for $\begin{pmatrix}0\\k\end{pmatrix}$ or $\begin{pmatrix}k\\2\end{pmatrix}$
4(b)(ii)	$\begin{pmatrix}-2\\9\end{pmatrix}$	**2**	**FT** *their* $\overrightarrow{PQ}$ **B1FT** for $\begin{pmatrix}0\\6\end{pmatrix}$
4(c)(i)	$\frac{2}{3}\mathbf{t}+\frac{1}{3}\mathbf{u}$ or $\frac{1}{3}(2\mathbf{t}+\mathbf{u})$ final answer	**2**	**M1** for $\overrightarrow{UY} = \frac{2}{3}(\mathbf{t}-\mathbf{u})$ oe or $\overrightarrow{TY} = \frac{1}{3}(\mathbf{u}-\mathbf{t})$ oe or correct route soi
4(c)(ii)	$\frac{2}{3}\mathbf{t}$ cao	**1**	
5(a)	[x =] 7 [y =] 3	**2**	**B1** for each
5(b)	[x =] 0, [y =] 2 [x =] –3, [y =] 5	**4**	**B3** for $x = 0$ and $x = -3$ or **B2** for $x^2 + 3x = 0$ or **M1** for $2 - x = x^2 + 2x + 2$ If 0 scored award **B1** for $x = 0$, $y = 2$ or $x = -3$, $y = 5$ from no/incorrect working ALTERNATIVE **B3** for $y = 2$ and $y = 5$ or **B2** for $y^2 - 7y + 10 = 0$ or **M1** for $y = (2 - y)^2 + 2(2 - y) + 2$ If 0 scored award **B1** for $x = 0$, $y = 2$ or $x = -3$, $y = 5$ from no/incorrect working